

which gives another necessary condition for (7) on the sphere S^d

$$(9) \quad \frac{1}{\beta} \geq d - \frac{2d}{q} - \sigma.$$

If the sharp Schrödinger admissible condition $\frac{1}{p} = \frac{d}{2}(\frac{1}{2} - \frac{1}{q})$ holds, then (9) is equivalent to

$$(10) \quad \frac{1}{\beta} \geq \frac{4}{p} - \sigma.$$

If the sharp wave admissible condition $\frac{1}{p} = \frac{d-1}{2}(\frac{1}{2} - \frac{1}{q})$ holds, then (9) is equivalent to

$$(11) \quad \frac{1}{\beta} \geq \frac{d}{d-1} \frac{4}{p} - \sigma.$$

Now, we introduce our main new results on the fractional Schrödinger, the wave and the Klein-Gordon equations on compact manifolds.

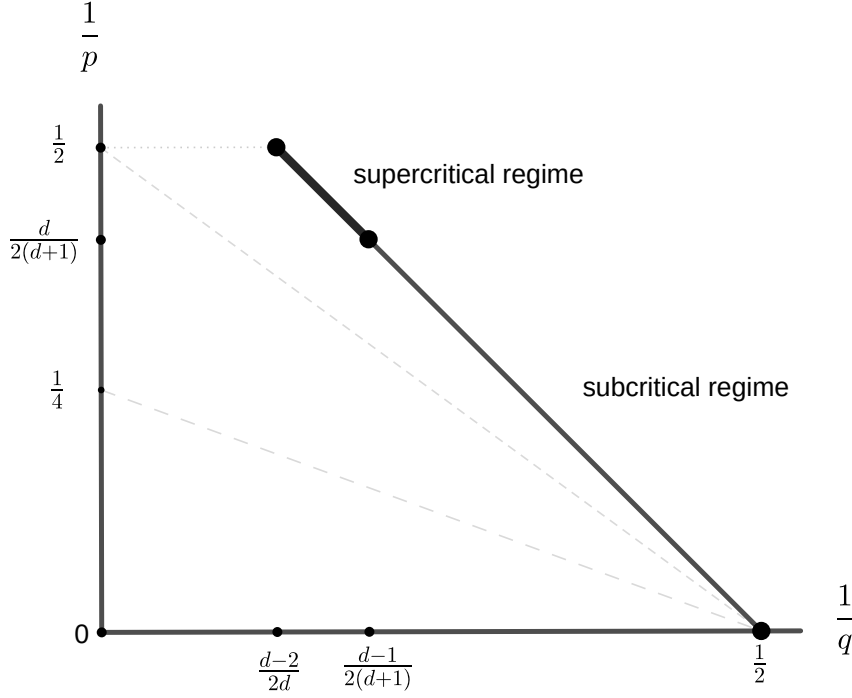


FIGURE 1. Sharp Schrödinger admissible pairs

1.2. Fractional Schrödinger equation. Suppose $p \geq 2$, $q < \infty$ and $\frac{1}{p} = \frac{d}{2}(\frac{1}{2} - \frac{1}{q})$. We divide these sharp Schrödinger admissible pairs (p, q) into four groups. See Figure 1.

- (i) Subcritical regime: $d \geq 1$, $2 \leq q < \frac{2(d+1)}{d-1}$
- (ii) Critical point: $d \geq 2$, $q = \frac{2(d+1)}{d-1}$
- (iii) Supercritical regime: $d = 2$, $\frac{2(d+1)}{d-1} < q < \infty$ or $d \geq 3$, $\frac{2(d+1)}{d-1} < q < \frac{2d}{d-2}$
- (iv) Keel-Tao endpoint: $d \geq 3$, $q = \frac{2d}{d-2}$.